

Collective Charge Current and Current-Correlation Specification

C0 Contract Closure and C1 Ordered Collective Correlations

mdstats

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Purpose and status

This document specifies roadmap stages **C0** and **C1** for `mdstats 0.19.85a0`:

- C0 closes the package-level charge, neutrality, grouping, correlation-order, volume-provenance, and immutability contracts.
- C1 implements collective charge-current construction and positive-lag total and ordered group-current correlations.

The implementation resides in `mdstats.analysis.current_correlation`. Conductivity integration, explicit plateau estimation, and the Nernst-Einstein comparison are implemented by the separate C2 module `mdstats.analysis.ionic_conductivity` in `mdstats 0.19.86a0`.

The equilibrium current-correlation framework follows the Green-Kubo linear response relation [1, 2]. FFT evaluation reuses the package's existing positive-lag linear-correlation machinery, which is an implementation of the Wiener-Khinchin correlation identity [3, 4]. Charge resolution, exact group partitioning, fixed/variable-cell provenance, public result schemas, backend selection, and fail-closed validation are `mdstats` design decisions.

Physical definitions

For per-atom charges q_i in units of the elementary charge e and Cartesian velocities $\mathbf{v}_i(t)$ in Angstrom/ps, the microscopic charge current is

$$\mathbf{J}_q(t) = \sum_i q_i \mathbf{v}_i(t).$$

Its stored units are

$$e \text{ Angstrom/ps}.$$

For an exact partition of the current-carrying atoms into named groups a ,

$$\mathbf{J}_q(t) = \sum_a \mathbf{J}_a(t).$$

The positive-lag ordered cross correlation is

$$C_{ab}(\tau) = \langle \mathbf{J}_a(t_0) \cdot \mathbf{J}_b(t_0 + \tau) \rangle_{t_0}.$$

At positive lag, C_{ab} and C_{ba} are distinct estimators and are stored separately. No implicit symmetrization is performed.

C0 resolved contract

Charge input

`compute_charge_current()` accepts exactly one charge source:

1. `charges`: one finite real value per canonical atom; or
2. `species_charges`: a mapping from exact chemical symbols to finite charge values.

Integer-key charge mappings are rejected because an integer could denote an atom index or an atomic number. A species mapping must contain every element present in the collection exactly once and must not contain unknown or unused symbols. Resolved charges are stored as an immutable per-atom array in canonical atom order.

Atoms with exactly zero resolved charge do not contribute to the current and are excluded from `current_atom_indices`. At least one current-carrying atom is required.

Neutrality

The first release requires a neutral system:

$$\left| \sum_i q_i \right| \leq \varepsilon_Q,$$

where `neutrality_tolerance_e` is finite and nonnegative. A non-neutral override is deliberately absent. This prevents an unqualified drift correction from changing the total charge current by $Q_{\text{tot}} \mathbf{v}_{\text{drift}}$.

Drift removal

Velocity validation and drift construction reuse `_velocity_common.py`. `drift_mode`, `drift_species`, and `drift_atom_indices` therefore have the same meaning and strict validation as VACF and direct velocity-spectrum analysis. The exact drift-reference atom population is retained in the shared `DynamicsInputSignature`.

For a neutral system, subtracting one common framewise drift velocity leaves the total current invariant analytically. Individual charged-group currents can change, so the selected drift convention remains part of the result provenance.

Exact group partition

`species_groups` is optional. When supplied, it is an insertion-ordered mapping from a nonempty unique group name to a `SpeciesSelection`.

Each group selection is intersected with `current_atom_indices`; zero-charge atoms are ignored. The resulting nonempty groups must be pairwise disjoint and must cover every current-carrying atom exactly once. Consequently,

$$\mathbf{J}_q(t) = \sum_a \mathbf{J}_a(t)$$

is a validated result invariant. Partial, overlapping, or diagnostic groupings belong to a future separately named API.

Cell and volume provenance

ChargeCurrentResult stores:

- the complete instantaneous volume series;
- the three periodic-axis flags;
- whether the full cell matrix is fixed across all frames; and
- the fixed volume when the cell is fixed.

A trajectory with constant determinant but changing cell matrix is classified as variable-cell. C1 does not reject variable cells because current correlation itself requires no volume. C2 conductivity integration will reject them in its first release.

Public APIs

Charge-current construction

```
def compute_charge_current(
    collection: AtomisticFrameCollection,
    *,
    charges: ArrayLike | None = None,
    species_charges: Mapping[str, float] | None = None,
    species_groups: Mapping[str, SpeciesSelection] | None = None,
    drift_mode: DriftMode | None = None,
    drift_species: SpeciesSelection = None,
    drift_atom_indices: ArrayLike | None = None,
    neutrality_tolerance_e: float = 1.0e-12,
) -> ChargeCurrentResult:
    ...
```

The function requires a uniformly sampled trajectory with complete Cartesian velocities. It performs no time correlation, SI conversion, conductivity integration, smoothing, or plateau selection.

Current correlation

```
def compute_current_correlation(
    current: ChargeCurrentResult,
    *,
    max_lag: int | None = None,
    origin_stride: int = 1,
    lag_stride: int = 1,
    compute_tensor: bool = True,
    backend: Literal["auto", "direct", "fft"] = "auto",
) -> CurrentCorrelationResult:
    ...
```

max_lag defaults to half the trajectory length. Lag zero is always retained. origin_stride != 1 requires the direct backend. lag_stride changes only the stored lags, not the physical sample spacing. C1 correlates the raw resolved current. It does not subtract a time-mean current, detrend, smooth, or symmetrize the data; drift removal must be requested during charge-current construction.

Result schemas

ChargeCurrentResult

The immutable result stores:

times_ps	(T,)
total_current	(T, 3)
group_names	tuple[str, ...]
group_currents	(T, G, 3) or None
charges_e	(N,)
current_atom_indices	(M,)
group_atom_indices	mapping[str, (M_g,)]

total_charge_e	scalar
sample_spacing_ps	scalar
pbcs	(3,)
cell_volumes_a3	(T,)
cell_mode	"fixed" or "variable"
fixed_volume_a3	scalar or None
signature	DynamicsInputSignature
metadata	recursively immutable mapping

The constructor validates finite values, shapes, exact current-carrying atom identity, neutrality, exact group partition, exact group-current sum, fixed-cell volume consistency, and signature consistency.

CurrentCorrelationResult

The immutable result stores:

lag_steps	(L,)
lag_times	(L,)
scalar	(L,)
components	(L, 3)
tensor	(L, 3, 3) or None
group_names	tuple[str, ...]
group_scalar	(L, G, G) or None
group_tensor	(L, G, G, 3, 3) or None
n_origins	(L,)
backend	"direct" or "fft"
charges_e	(N,)
current_atom_indices	(M,)
group_atom_indices	immutable mapping
pbcs	(3,)
cell_volumes_a3	(T,)
cell_mode	"fixed" or "variable"
fixed_volume_a3	scalar or None
signature	DynamicsInputSignature
metadata	recursively immutable mapping

The scalar and Cartesian identities are

$$C_{\text{tot}}(t) = \sum_{\alpha} C_{\alpha\alpha}(t),$$

and, when groups exist,

$$C_{\text{tot}}(t) = \sum_{a,b} C_{ab}(t).$$

If tensors are retained, components equals the total tensor diagonal and each group_scalar[:,a,b] equals the trace of group_tensor[:,a,b,:].

Direct estimator

For lag k and valid origins $t_0 = 0, s, 2s, \dots < T - k - 1$, the direct estimator computes

$$C_{\alpha\beta}(k) = \frac{1}{N_k} \sum_{t_0} J_{\alpha}(t_0) J_{\beta}(t_0 + k).$$

Group tensors replace the two total currents with $\mathbf{J}_a$ and $\mathbf{J}_b$. The origin count is

$$N_k = \left\lfloor \frac{T - 1 - k}{s} \right\rfloor + 1.$$

FFT estimator

For origin_stride == 1, C1 forms zero-padded linear cross spectra

$$\hat{C}_{xy} = \hat{x}^* \hat{y}$$

and inverts them through `_fft.positive_lag_correlation_from_spectrum`. Division by the exact pair count $T - k$ occurs before lag subsampling. Group pairs are inverted one ordered pair at a time so temporary memory does not scale as $G^2 \times 3 \times 3 \times N_\omega$.

The FFT and direct estimators must agree within floating-point tolerance and use the same tensor orientation: the first Cartesian index belongs to the origin current and the second to the lagged current.

Backend selection

`backend="auto"` compares conservative direct and FFT work estimates. It keeps the direct estimator for short trajectories and whenever `origin_stride != 1`. The selected backend and both work estimates are stored in metadata.

Units

Current units are

`e*Angstrom/ps`

Current-correlation units are

`e^2*Angstrom^2/ps^2`

C1 does not convert the elementary charge, Angstrom, or picosecond to SI units. That conversion belongs exclusively to C2 conductivity integration.

Failure policy

The public functions fail closed for:

- missing or multiple charge sources;
- non-finite, wrong-shaped, ambiguous, incomplete, or extra charge mappings;
- a system with no nonzero charges;
- non-neutral total charge outside tolerance;
- invalid drift selections;
- empty, overlapping, or incomplete current groups;
- nonuniform time sampling or missing velocities;
- invalid lag, stride, boolean, or backend inputs;
- FFT requests with non-unit origin stride;
- malformed result construction;
- non-finite current or correlation values; and
- any violation of total/group or tensor/trace identities.

Required tests

C0-C1 acceptance requires tests for:

1. exact per-atom array and species-map charge agreement;
2. missing, duplicate-source, incomplete, extra, integer-key, and non-finite charge rejection;
3. neutrality-tolerance acceptance and non-neutral rejection;
4. neutral rigid translation giving zero total current;
5. explicit paired-charge algebra;
6. exact group-current sum;
7. overlap, omission, and empty-group rejection;
8. fixed-cell and constant-volume variable-cell classification;
9. drift-reference signature preservation;
10. direct/FFT agreement for total and group tensors;
11. ordered positive-lag group correlations with $C_{ab} \neq C_{ba}$;
12. exact total/group-pair sum and tensor-trace identities;

13. zero-current behavior and constant nonzero-current preservation;
14. strict integer and boolean validation;
15. backend and origin-stride policy;
16. deep array, mapping, and nested-metadata immutability;
17. public exports; and
18. regression compatibility with the existing VACF/dynamics branch.

Implemented C2 boundary

mdstats 0.19.86a0 consumes `CurrentCorrelationResult` in the separate `ionic_conductivity.py` module. C2 enforces full three-dimensional periodicity and fixed full-cell-matrix provenance, performs cumulative Green-Kubo quadrature with exact SI conversion, retains ordered group-pair contributions, selects only explicit conductivity plateau intervals, and compares against compatible full-3D species diffusion estimates. The C0-C1 result schemas remain the authoritative charge, partition, cell, and trajectory provenance source.

References

- [1] M. S. Green, “Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II. Irreversible Processes in Fluids,” *Journal of Chemical Physics* **22**, 398-413 (1954). DOI: 10.1063/1.1740082.
- [2] R. Kubo, “Statistical-Mechanical Theory of Irreversible Processes. I. General Theory and Simple Applications to Magnetic and Conduction Problems,” *Journal of the Physical Society of Japan* **12**, 570-586 (1957). DOI: 10.1143/JPSJ.12.570.
- [3] N. Wiener, “Generalized Harmonic Analysis,” *Acta Mathematica* **55**, 117-258 (1930). DOI: 10.1007/BF02546511.
- [4] A. Khintchine, “Korrelationstheorie der stationären stochastischen Prozesse,” *Mathematische Annalen* **109**, 604-615 (1934). DOI: 10.1007/BF01449156.